

Ahmed Mahmoud Abdelaal

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[GitHub](#) | [LinkedIn](#)

Education

Faculty of Engineering - Cairo University

Bachelor's degree in system and biomedical engineering

2021 – present

Experience

Backend Mentor

Jan – present 2025

IEEE EMBS Student Branch Chapter – Cairo University

Mentored students in backend development technologies, Conducted workshops and hands-on sessions on Node.js, MongoDB, Express.js.

Fullstack Developer

Oct 2024

NASA SpaceApps Hackathon – Exosky Project

Developed Exosky, a web application exploring exoplanets and stars using NASA APIs, Implemented 3D visualizations, search functionality, a drawing constellations mode.

Courses and Certificates

- Back End Development and APIs [freeCodeCamp](#)
- SQL – Certificate [Hackerrank](#)
- REST – Certificate [Hackerrank](#)
- Getting Started with Deep Learning [Nvidia](#)
- Building Real-Time Video AI Applications [Nvidia](#)
- IBM – Machine Learning with Python [Coursera](#)
- Supervised Machine Learning – DeepLearning.AI [Coursera](#)

Technical Skills

- **Web Development:** REAT API, Javascript, HTML5, CSS, Nodejs, Expressjs, MongoDB, PostgreSQL, Jest, Reactjs, TailwindCSS, Redux, Material UI, Docker
- **Application Development:** Python, PySide6, PyQt5, Matplotlib, C++, OOP
- **Deep learning :** Pandas, NumPy, scikit-learn, Keras, Pytorch
- **Tools & Libraries:** Git, Socket.io, Bash script

Languages

- **Arabic:** Native/Fluent
- **English:** Intermediate

Projects

Learnova In progress - [GitHub](#)

Educational platform for interactive learning Developed a scalable web application to facilitate online learning

- Integrated features for course management, quizzes, and user progress tracking.
- Utilized modern web technologies for a responsive and engaging user experience.

Redis-monitor-dashboard

Jun 2025 - [GitHub](#)

A real-time Redis monitoring dashboard

- Real-time Monitoring: Live Redis metrics with WebSocket updates.
- Interactive Charts: Memory usage, operations per second, connected clients.

Sociopedia – Social Media Platform

Feb 2024 – [GitHub](#)

Dynamic full-stack social media platform

- Built using Node.js, Express.js, React.js, Material-UI, and MongoDB.
- Implemented JWT-based authentication for secure user access.
- Utilized Redux for efficient state management.

ChatEase – Real-Time Chat Application

Aug 2024 - [GitHub](#)

Full-featured messaging platform with real-time capabilities

- Developed using React, Node.js, Socket.io, and MongoDB.
- Implemented JWT-based authentication, real-time messaging, and media uploads.
- Created responsive UI with Tailwind CSS and seamless communication with Socket.io.

Skin-Cancer-Detection

May 2025 - [GitHub](#)

Computer vision and ML pipeline for skin cancer detection from dermoscopic images

- Apply preprocessing techniques to enhance image quality.
- Extracts texture and color features using traditional computer vision methods.
- Trains ensemble classifiers to differentiate between malignant and benign.

FilterLab - Digital Filter Designer

Jan 2025 - [GitHub](#)

- Built using PyQt5 and Matplotlib for interactive filter design.
- Supports standard filters and all-pass filters with visualization.
- Generates corresponding C code for designed filters.

TuneSpy

Jan 2025 – [GitHub](#)

Python application for audio analysis and fingerprinting

- Generates spectrograms and extracts MFCC features from audio files.
- Compare loaded audio with a preprocessed database to find similar matches.

Sampling-Studio

Jan 2025 - [GitHub](#)

Interactive tool exploring the Nyquist–Shannon sampling theorem

- Features real-time signal sampling, reconstruction, and frequency analysis.
- Tools for signal mixing, noise control, and comparison of reconstruction methods.